

Research and Writing Skills for Academic and Graduate Researchers

RESEARCH AND WRITING SKILLS FOR ACADEMIC AND GRADUATE RESEARCHERS

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ACKNOWLEDGEMENT OF COUNTRY

In Aboriginal and Torres Strait Islander culture it is respectful to formally womin djeka (welcome) people to Aboriginal and Torres Strait Islander country or land. At RMIT we do this with the following acknowledgement:



'Luwaytini' by Mark Cleaver,
Palawa.

present.”

“We would like to acknowledge Wurundjeri people of the Kulin Nations as the Traditional Owners of the land on which the University stands. We respectfully recognise Elders past and

At RMIT we recognise and respect the unique culture and contribution that Aboriginal and Torres Strait Islander people bring to our communities. We are also proud to provide study, cultural, and personal support to our Aboriginal and Torres Strait Islander students in their learning journey. We womin

djeka you to further explore and experience more about Australian Indigenous culture by visiting our Indigenous education centre – Ngarara Willim Centre and interacting with our Aboriginal and Torres Strait Islander staff and students.

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- It has been optimised for people who use screen reading technology.
 - All content can be navigated using a keyboard.
 - Links, headings, and tables are formatted to work with screen readers.
 - Images have text alternatives.
- Information is not conveyed by colour alone.

Other file formats available

In addition to the web version, this book is available in a number of file formats, including:

- **Print PDF:** the format you should select if you plan to make a printed, physical copy of your book. This file meets the requirements of printers and print-on-demand services.
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- **and various editable files:** look for the 'Download this book' drop-down menu on the landing page to select the file type you want.

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While we strive to ensure that this resource is as accessible and usable as possible, we might not always get it right. We are always looking for ways to make our resources more accessible.

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This page provides a record of edits and changes made to this text from pre-publication to its latest iteration. Whenever edits or updates are made in the text, we provide a record and description of those changes here.

If you have a correction or recommendation you would like to suggest, please contact the RMIT University Library Research Services Team.

Version	Date	Type
Pre-publication	2017-2022	Graduate Researcher KnowHow
Pre-publication	2018-2022	Research Skills for HDR Candidates
V1	21/09/2022	Research and writing skills for academic and graduate researchers
V2	16/12/2022	Research and writing skills for academic and graduate researchers
V3	15/02/2023	Research and writing skills for academic and graduate researchers

ABOUT THIS BOOK

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Artwork 'Luwaytini' by Mark Cleaver, Palawa (underlayed)

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RESEARCH AND WRITING SKILLS MODULE HUB

These stand-alone modules on research and writing skills can be completed individually and should take around 30 minutes, maybe a little longer if you do all the exercises. Work through any module systematically, or browse to find relevant material using the contents menu on the left hand side.



Literature searching and managing references



Research writing and study skills



Publishing and metrics

This set of modules is under development –
check back regularly to find additional content.

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concentric circles – Photo by Jordan McDonald on Unsplash

STRATEGIES AND RESOURCES FOR SEARCHING THE LITERATURE

Welcome to this online module related to strategies and resources for searching the literature. The module will explore key resources to use when literature searching, and the features of developing an effective search strategy.

Learning objectives

This module will introduce you to:

- strategies and search techniques that can be used in your searches
- how to construct your search query

- selecting key resources to use in your searches, such as LibrarySearch, Databases, Google Scholar
- staying current with the literature.

There are learning activities throughout this module to help you with some practical searching.

This module should take you about 30 minutes to complete.

Work through each section using the navigation footer (i.e. previous/next), or use the contents menu to select a particular section.

Upon completion, please help us improve this module by taking a moment to

complete the feedback form provided at the end.

You may also be interested in viewing the additional resources about Literature searching in the Explore Further chapter at the end of this module. Provided is a webinar



Photo by Erik Mclean on Unsplash

recorded from the Library's 'PhD Up!' program (now called Research Plus) and associated library guides.

4 | STRATEGIES AND RESOURCES FOR SEARCHING THE LITERATURE

BEING CLEAR ABOUT YOUR PURPOSE

The purpose

When considering how and where to start your literature search, it is first worth thinking about why you are looking for information. What is the purpose of your search?

You may be doing a literature search:

- for a research proposal
- for a literature review as part of a thesis or manuscript
- for an assignment in an academic course
- to update your knowledge
- for professional development
- to support a workplace activity.

Determining the reason for your literature search will help you select the types of materials needed. The Library has a wide range of materials available including books, journals, videos, newspapers, images, standards, case studies, theses, and more. This module will introduce you to some of these types of materials. To explore these further go to Library collections.

What is a literature review?

Test your knowledge



[https://rmit.pressbooks.pub/
researchwritingmodules/?p=732#h5p-8](https://rmit.pressbooks.pub/researchwritingmodules/?p=732#h5p-8)

There are two main purposes for undertaking a literature review.

1) To show awareness of the present state of knowledge in a particular field, including:

- seminal authors
- the main empirical research
- theoretical positions
- controversies
- breakthroughs as well as links to other related areas of knowledge.

2) To provide a foundation for the author's research. The literature review should:

- help the researcher define a hypothesis or a research question, and how answering the question will contribute to the body of knowledge
- provide a rationale for investigating the problem and the selected methodology
- provide a particular theoretical lens, support the argument, or identify gaps.

This video provides an overview of what a literature review is. Watch this video and answer the question below.

Literature Reviews: An Overview for Graduate Students (9:39 mins)



One or more interactive elements has been excluded from this version of the text. You can view them online here:
<https://rmit.pressbooks.pub/researchwritingmodules/?p=732#oembed-1>

“Literature Reviews: An Overview for Graduate Students” by libncsu is licensed under CC BY-NC-SA 3.0

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

*[https://rmit.pressbooks.pub/
researchwritingmodules/?p=732#h5p-4](https://rmit.pressbooks.pub/researchwritingmodules/?p=732#h5p-4)*

Further information and resources on writing your literature review can be found in the Learning Lab.

PLANNING YOUR SEARCH

A literature search starts with effective planning. A literature search plan can:

- ensure your literature searching is comprehensive
- prompt you to think critically
- guide your searching to focus on your topic
- document your search processes to minimise replication
- save time.

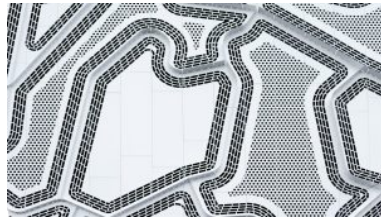


Photo by [Ussama Azam](#) on [Unsplash](#)

Identify the main or important aspects

It can be helpful to write down your research question and identify the important words that define your topic before searching.

Example

‘I am interested in the recycling of waste water in cities.’

The important terms to include for this topic would be:

- **recycling**
- **waste water**
- **cities**

Developing keywords

The next step is to compile a list of synonyms or alternative terms. This will ensure that you will have a comprehensive search for relevant literature.

Consider variations such as:

- synonyms or alternative terms – e.g. climate change, global warming

- acronyms / abbreviations – e.g. DFT, density functional theory
- alternative Spellings – e.g. aluminium, aluminum
- plurals / alternative endings – e.g. environment, environmental.

For our research topic, here is the start of a list of synonyms and alternative terms.

Recycling	Reuse
Waste water	Sewerage
Cities	Urban

Now look to developing your search strategy utilising some different search techniques.

DEVELOPING A SEARCH STRATEGY

There are a number of different search techniques that you can use to develop your search strategy. This chapter explains how to make your searching thorough and effective by using phrase searching, truncation, spelling variations, and Boolean operators.

Phrase searching

To specify that two or more words must appear as a phrase, use double quotes.

Example

"global warming"

Truncation

To search for alternative endings of words, use truncation or stem searching. In most databases, the truncation operator is an asterisk (*).

Example

A search for **environment*** will retrieve:

- environment
- environments
- environmental
- environmentalism

Spelling variations

Sometimes there are slight differences in the spelling of a word, for example, English and American variations. To facilitate searching on alternative spellings, some databases allow

wildcard searching. Wildcards **take the place of one or more characters in a search term**. The wildcard operator is often a question mark (?) or an asterisk (*).

Example

A search for **colo?r** will find:

- color
- colour

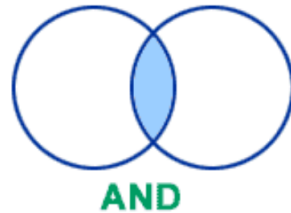
Once you have a list of keywords, synonyms, and effective search techniques you will then need to combine these into effective search statements.

Combining with Boolean

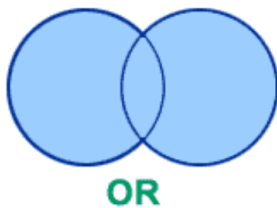
Boolean searching can be very helpful when searching databases. It allows you to combine multiple terms with connecting words, called 'Boolean operators'. Most databases use common Boolean operators, such as AND, OR, and

NOT. Here are examples of how you can use Boolean operators to broaden or narrow your search.

AND operator narrows your search

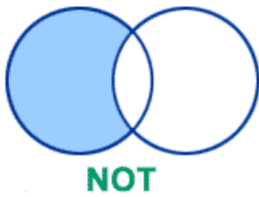


Use AND operator to find records that contain all the terms you use. This is useful if you want to find only those records that mention both terms. Searching for “climate change” AND cereals will locate records with both these terms.



OR operator broadens your search

Use OR operator to find records that contain any of the terms you use. This is useful if you want to allow for alternative terms/synonyms for the same concept. Searching for “climate change” OR “global warming” will locate records with both as well as one of these terms.



NOT operator narrows your search

Use NOT operator to exclude a term. This is useful if you don't want to find any records that contain a particular term. Use it with care to avoid excluding relevant articles that briefly mention the second term. Searching for “global warming” NOT “fossil fuel” will exclude all records that mention “fossil fuels” as well as “global warming” and “fossil fuels” together.

Watch this video to gain an understanding of how to use Boolean operators.

Boolean Operators (3:02 mins)



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=549#oembed-1>

Boolean Operators (3:02 min) by Lexy Spry and Emily Wixson (YouTube)

Test your knowledge



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=549#h5p-7>

THE SEARCH QUERY

To construct the search query consider using some search techniques such as Boolean operators and phrase searching.



Photo by [Justin Ha](#) on [Unsplash](#)

In considering the topic **I am interested in the recycling of waste water**

in cities, three main aspects were identified, and a list of possible keywords (search terms) was compiled.

The search statement may look like the following:

Example

(recycling OR reuse) AND (“waste water” OR

sewerage OR “drain water”) AND (cities OR urban)

This search query has used:

- double quote marks enclosed around keywords that contain two words so they are searched as a phrase
- Boolean operator OR is used to combine keywords about the same aspect
- Boolean operator AND is used to combine keywords about different aspects.

The search features of library databases make constructing your search query easier, especially when using advanced search options. For example, the Boolean operators can usually be selected from a drop-down menu. This is evident in the example below, showing the search query where the Boolean operators of AND are selected.

Note: nearly all databases have these features, but they might look different. You may need to explore to find the options you need.

Database

Advanced Search

recycling OR reuse

Anywhere

AND

"waste water" OR sewage OR "drain water"

Anywhere

AND

cities OR urban

Anywhere

[+ Add a row](#)
[- Remove a row](#)

Limit to:
☐ Full text
☒ Peer reviewed

Publication date:

Last 3 years

Search

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Activity

Now it's your turn to develop your search query for your own research topic.

1. Write down your research topic and identify your main aspects.
2. For each aspect list some alternative search terms (synonyms).

3. Consider some different search techniques that you may need to use.
4. Create a search query that combines your topic keywords and synonyms using Boolean operators AND and OR.

REFINING YOUR SEARCH

After developing your search query and executing your search, you now need to look at the search results and determine whether you have been successful in locating the desired literature. Maybe you need to refine your search further. Outlined below is information about using the databases fields or the subject headings to return more accurate search results.

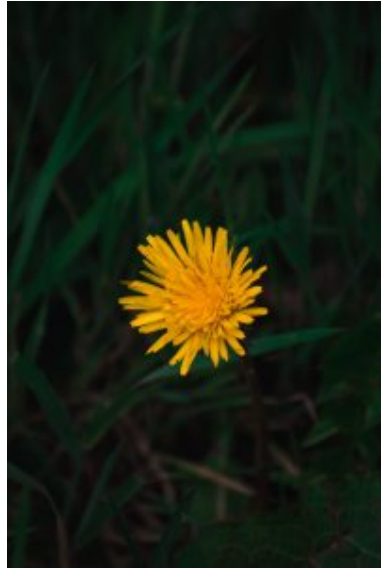


Photo by [Shyam on Unsplash](#)

The limits options of the database are also another way to refine your search results.

Field searching

It is the advanced search option in a database that provides

searching by a particular field. If the search terms appear in the title or subject of the reference, then it is likely to be more relevant.

References in a database are indexed by different fields, for example:

- Author
- Title
- Journal name
- Subject.

Subject headings

Some databases will have subject headings, or descriptors, that are assigned to the reference when it is indexed in the database. The subject headings are used to describe the content of the reference. This list of subject headings is called a thesaurus. The thesaurus may have an elaborate structure of broader, narrower, and related terms, which will help you to search very precisely. Check the database's help for information about searching by subject headings.

Limits

Many databases will give you the option to limit your results. Look for the different limit options in the database when you are getting too many results.

For example, you may be able to specify that you want to retrieve by:

- Material format (e.g. peer-reviewed journals)
- Publication date (e.g. last five years)
- Language (e.g. English only)

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=737#h5p-9>

SELECTING RESOURCES TO SEARCH

Library subject guides

The Library's subject guides are good starting points for research. They list relevant databases and other resources relevant for different subject areas.

Watch this video for a brief introduction to '*What's a Library Subject Guide?*'

What's a Library Subject Guide? (1:13 mins)



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can view them online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=102#oembed-1>

What's a Library Subject Guide? (1:12 mins) by RMIT University (YouTube)

The following image shows how to locate Subject Guides from the Library homepage.

Find books, articles and more

[Library Search](#) [Databases](#) [Google Scholar](#) [Research Repository](#) [Not sure where to start?](#)

Library subject guides

Need help getting started with your research? These guides highlight the variety of resources you can access, including databases, e-journals, e-books and websites.

Select your study area:

- [Architecture and Urban Design](#)
- [Business and Law](#)
- [Engineering](#)
- [Property, Construction and Project Management](#)
- [Art](#)
- [Design](#)
- [Fashion and Textiles](#)
- [Science](#)
- [Biomedical, Health and Nursing](#)
- [Education](#)
- [Media and Communications](#)
- [Social Sciences and Languages](#)

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Topic-specific guides

Explore topic-specific guides such as:

- Industry information
- Company information
- Standards
- EndNote: a beginner's guide

Research guides

There is also a range of Research guides that include:

- Altmetrics
- Finding and obtaining theses
- HDR students
- Research Data Management
- Research Metrics
- Researcher Profiles and ORCID iDs
- Starting your literature review
- Staying current with your research
- Strategic Publishing
- Systematic Reviews.



Photo by [David Clode](#) on [Unsplash](#)

Activity

1. Go to the Library homepage and find a subject guide relevant to your research area.
2. Does that subject guide list any databases that might be useful for your research topic?

USING SEARCH ENGINES

Google and Google Scholar

Many people begin their research with Google, and sometimes Google Scholar. Google can be useful to get an overview, background information, or to define terms. We all Google – but are you doing it effectively?



Photo by Mitchell Luo on Unsplash

Using Advanced Google or Google Scholar improves your chance of obtaining relevant information. Researchers are expected to use scholarly information and Google alone is insufficient.

Google Advanced lets you choose combinations of words – e.g. all these words, exact phrase or word etc., and narrow results by language, region, last update, site or domain, file type, etc.

The following table provides a comparison of the difference in features between Google and Google Scholar.

Google vs. Google Scholar

Features	Google	Google Scholar
Easy to search interface	Yes	Yes
Coverage	Many sources, good definitions, background information, government/company reports, but uncontrolled content.	Indexes scholarly sources, but with uneven discipline content.
Format	Varied, news, company reports, trade and government publications, grey literature.	Scholarly material, articles, conference papers, book chapters, patents.
Search results with basic filtering	Yes, with Advanced Google.	Limited
Search alerts	No	Yes
Search metrics	No	Yes
Author profiles	No	Yes
Full-text access	Yes, depending on sources.	Yes, but payment may be required.
Export citations to reference managers	No	Yes

Best Practice Tip

Sign in to Google Scholar via the Library webpage to access full text that RMIT holds in linked databases.

See the image below for where to find Google Scholar on the RMIT Library homepage.

Find books, articles and more

[Library Search](#) [Databases](#) [Google Scholar](#) [Research Repository](#) [Not sure where to start?](#)

Google Scholar

Google Scholar search results include resources from our Library's online collections.

Access via the Library >

Easy access to full text articles from our Library collections. If accessing off-campus, you will need to log in with your RMIT password.

Direct access >

With access to personal Google account that allows exporting citations to Endnote and other bibliographic management tools.

Access information and tips >

We explain the two access options in more detail, give you some tips on how to interpret search results and what to do when there are access issues.

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Activity

Search Google Scholar using some of your search query keywords.

As you search make note of the following:

1. The different types of resources in your results list
2. Can you follow a link to the full text of the article or resource?
3. Look for where to refine your search by date or relevance, and also where you can include (or exclude) patents and citations

There are also further tools in Google Scholar to find related material. These are located under each of the titles listed in your search results.

Can you find the Cite, Cited by, Related articles, and Save links?

See if you can also find where to save an alert for your search.

Saving an alert will provide an email feed for any new papers on your topic, as they are added to Google Scholar.

Test your knowledge



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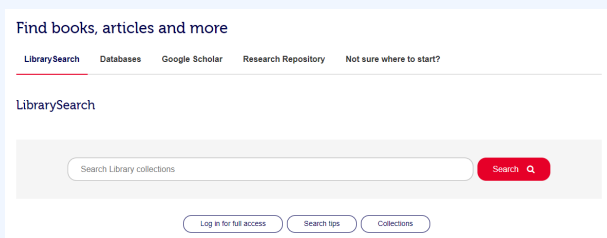
LibrarySearch

LibrarySearch is RMIT University Library's search engine and an important place to continue your literature search.

You can use LibrarySearch to retrieve a wide variety of materials:

- Books and E-books
- Journal and newspaper articles
- DVDs and streaming video
- Theses and more

Make sure to log in to LibrarySearch as some resources are only available when you log in. Look for the ‘Log in for full access’ button on the Library’s homepage



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Watch this video for a brief introduction to LibrarySearch

What is LibrarySearch? (1:07 mins)



One or more interactive elements has been



*excluded from this version of the text. You
can view them online here:
[https://rmit.pressbooks.pub/
researchwritingmodules/?p=149#oembed-1](https://rmit.pressbooks.pub/researchwritingmodules/?p=149#oembed-1)*

What is LibrarySearch? (1:07 mins) by RMIT University (YouTube)

LibrarySearch is particularly good for:

- Getting an overview of the Library’s physical and online collections
- Finding “something” on a topic
- The refine features to narrow your search results

There are refine options in the menu on the left,
including:

- Peer-reviewed articles, full text online
- Resource type – articles, books, etc.
- Subject
- Date range

To access full text for online material e.g. e-books or journal articles, you click on the ‘View Online’ link at the item record.

LibrarySearch does not provide even coverage of the subject area. Specialised subject databases in business, architecture, psychology, legal, and fashion are not covered by LibrarySearch.

As a researcher, you will also need to search specialist library databases. We will be looking at these databases later in this module.

Activity

Go to LibrarySearch on the Library homepage and search using keywords for your own research topic.

Locate a peer-reviewed article on your topic and access the full text.

Research Repository / Theses

The RMIT Research Repository is an open-access institutional repository providing free, searchable access to research publications authored by RMIT University staff and students.

Most theses submitted at RMIT University can be found in the RMIT Research Repository.

Theses can be invaluable sources for in-depth and significant research related to your field of study. The RMIT Finding and obtaining theses research guide is a great starting point for resources that will allow you to access Australian and international theses.

Activity

- Search the RMIT Research Repository
- Enter your search term, for example, climate change

- Limit your search results to 'Dissertations & Theses'
- The 'Refine your results' enables further selection of:
 - Resource type
 - Research unit
 - Author
- Identify a relevant thesis to your topic

SEARCHING LIBRARY DATABASES

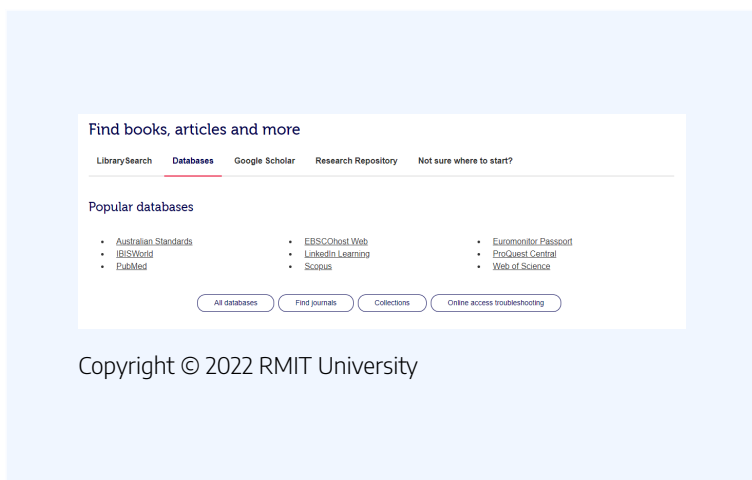
Databases

While Google Scholar and LibrarySearch can provide good starting places, you will need to move beyond these to comprehensively retrieve relevant literature.

Databases are excellent tools for researching journal articles, as well as other scholarly content. They often have a specialised focus – for example, by subject, resource type, or geography. You can explore the different types of databases on offer from the Library that include videos, images, theses, standards, and more.

There are many databases available, some of which are subject-specific. To locate databases relevant to your research area you can consult the Library's subject guides.

To access the list of popular databases, select the Databases tab in the 'Find books, articles and more' or the Library's homepage.



Watch this video for a brief introduction to Library databases.

What's a library database? (1:56 mins)



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=108#oembed-1>

What's a library database? (1:56 mins) by RMIT University (YouTube)

Test your knowledge



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=108#h5p-6>

Searching databases

The search results in databases can be sorted and filtered in a variety of ways – for example, by the level of content (scholarly, peer-reviewed).

A large database that covers a range of subject areas including business, science, health, and social sciences is ProQuest.

Finding the full article

Databases will often give you direct access to the full text online, but if you cannot see a link to the online article, try the 'Find it' button link. This will open a new browser window that will search all databases to which RMIT University Library has access.

If not held at RMIT

If you cannot find the article online it may be available in print, or you can request the article via the Library's Document Delivery Service.

Watch this video to learn how to do an advanced search in the ProQuest database.

ProQuest Advanced Search (2:14 mins)



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=108#oembed-2>

Activity

The activity uses the ProQuest database. You may wish to practice searching one of the databases you selected earlier, from your subject guide.

1. Go to the Library homepage and select the Databases link.
2. Select ProQuest Central from the 'Popular databases' list.
3. Note the Basic and Advanced Search options. Select Advanced Search.
4. Search the database using your search query keywords and try to use a simple Boolean search e.g. (waste water OR effluent) AND (recycling AND cities).
5. Next to the search box from the drop-down menu select 'Anywhere except full text'. Also, select options to limit to peer-reviewed articles and publication date of 'Last 3 years'.

6. Note options for Source/Document type.
7. Select search.

Citation databases

Scopus and Web of Science are two large citation databases that can be important starting points for subject areas, particularly in the sciences.

These databases also display citation data that enables you to track the development of a research idea forward through time.

Further information on searching these is below.

Watch this video to learn how to do a basic search in the Scopus database.

Scopus Tutorial: How to Conduct a Basic Search (2:57 mins)



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[https://rmit.pressbooks.pub/
researchwritingmodules/?p=108#oembed-3](https://rmit.pressbooks.pub/researchwritingmodules/?p=108#oembed-3)

Scopus Tutorial: How to Conduct a Basics Search (2:57 mins) by Scopus (YouTube)

Watch this video to learn how to do a search in the Web of Science database.

Web of Science: Search Tips (5:38 mins)



One or more interactive elements has been excluded from this version of the text. You can view them online here:

[https://rmit.pressbooks.pub/
researchwritingmodules/?p=108#oembed-4](https://rmit.pressbooks.pub/researchwritingmodules/?p=108#oembed-4)

Web of Science: Search Tips (5:38 mins) by Web of Science (YouTube)

STAYING CURRENT WITH THE LITERATURE

Another useful feature of different databases includes the ability to set up ‘search alerts’, which allows you to receive alerts of new articles that meet your search criteria. This is an excellent way to keep up-to-date with your research.

It is also possible to set up alerts for:

- The latest tables of contents for specific relevant journals.
- Citation alerts whenever a particular article is cited by someone.
- Upcoming conferences and calls for papers.
- Websites, social media, and Google.

See the library guide to Staying current with your research for more details and examples.

Conferences

Conferences are an important part of staying current with intellectual and academic developments and discussions. There are diverse websites that will alert you of upcoming conferences. Papers are sometimes available via conference websites or accessible via databases.



Rice University Stock Photography by [Ed Schipul](#) (CC-BY-3.0)

The Staying Current With Your Research Library Guide lists a number of conference directories.

EXPLORE FURTHER

Congratulations on completing this module. We have explored the purpose of a literature review, some of the key resources for searching the literature, and some aspects of developing a search strategy.



Photo by [Joseph Chan](#) on [Unsplash](#)

The module also included information on staying current with new literature published in the area and searching for theses and conferences.

Further assistance

For more assistance in researching your literature review contact the Library by submitting a research request form.

Additional information and resources are available in the guides:

- Starting your literature review
- Staying current with your research
- Finding and obtaining theses

Information and resources on writing your literature review can be found in the Learning Lab.

Learn more

You may also be interested in viewing the following webinar Strategies and resources for searching the literature (64 mins.) recorded from the Library's PhD Up! program (now called Research Plus). The webinar content complements this online module.

Feedback

Your comments and suggestions on how we can improve this module will be appreciated. Please complete our feedback form.

What's next?

You have completed this module on Strategies and Resources for Searching the Literature. You can return to the Research and Writing Hub to choose another module to complete.

CONDUCTING A SYSTEMATIC REVIEW



Photo by Florian Schmetz on
Unsplash

Welcome to this online module on how to conduct a systematic review. The module provides an overview of the key components and the process you need to follow when conducting a systematic review. Briefly covered are

developing the research question and the protocol, aspects of searching, performing the analysis, and reporting findings.

Learning objectives

The module will help you:

- understand what a systematic review is
- know the difference between a systematic review and a literature review
- know the various steps involved in conducting a systematic review
- develop your research question and protocol
- learn about resources and strategies when searching the literature
- to understand the selection and evaluation processes of the included studies.

The course materials include information, videos, and learning activities that explore key features of conducting a systematic review.

This module should take you about 30 minutes to complete.

Work through each section using the navigation footer

(i.e. previous/next), or use the contents menu to select a particular section.

Upon completion, please help us improve this module by taking a moment to complete the feedback form provided at the end.

You may also be interested in viewing the additional resources about Systematic Reviews in the Explore Further chapter at the end of this module. Provided is a webinar recorded from the Library's 'PhD Up!' program (now called Research Plus), a short video introducing systematic reviews, and the associated library guide.

Please note that the examples used in this module are related to the discipline of health sciences, but the principles of this research methodology may be applied to other areas.

ABOUT SYSTEMATIC REVIEWS

A systematic review is a secondary research method that identifies and evaluates evidence from existing data in primary research studies.

Commonly, systematic reviews are used in healthcare research to assess the evidence on whether a medical intervention is effective in treating a certain condition.



Photo by [José Ignacio González Pansiera](#) on [Unsplash](#)

Definition

As defined by the Cochrane Collaboration a systematic review is:

“A review of a clearly formulated question that uses systematic and explicit methods to identify, select, and critically appraise relevant research, and to collect and analyse data from the studies that are included in the review. Statistical methods (meta-analysis) may or may not be used to analyse and summarise the results of the included studies.”

Features of a systematic review

A systematic review has these key features:

1. It addresses a **research question** that is clearly defined and specific.
2. It follows a **rigorous methodology** by adopting a clearly **pre-defined protocol**.
3. It is an **exhaustive search** of the literature that is carried out in accordance with the protocol.
4. All evidence is considered and only excluded if it does not meet the **eligibility criteria**.

5. The included studies are **critically analysed**, especially for the risk of bias.
6. It ideally involves **more than one person**.

For HDR candidates

A PhD or a Masters by Research thesis will include a chapter devoted to a review of the literature. This type of review is known as a traditional or narrative review, or simply a literature review.

The following table outlines the different features of a systematic review and a traditional literature review.

Systematic review vs. Literature review

Features	Systematic review	Literature review
Aim	Tightly specified objectives to answer a specific research question	Gain a broad understanding and description of a field
Scope	Narrow focus	Big picture
Planning the review	Transparent process with documented audit trail defined in a protocol	Nothing defined, allows for creativity and exploration
Searches	Rigorous and comprehensive search for ALL studies, explicit search strategy across numerous sources	Searching is probing, moving from study to study, following-up leads
Study selection	Predetermined criteria for including and excluding studies uniformly applied	Selection is variable as determined by the reviewer
Appraisal	Checklists to assess the quality of studies	Based on the reviewer's opinion
Synthesis	Tabular format with short summary answers	Discursive
Methodology	Must be presented for transparency	Not necessarily provided
Inferences	Based on all available evidence	Based on a sample of the evidence
Timeline	Months to years (average 18 months)	Weeks to months
Authors	Three or more	One or more

Features	Systematic review	Literature review
Value	Connects practising clinicians to high-quality evidence; Informs evidence-based practice	Provides a summary of literature on a topic

Watch the following short video to learn about the difference between a systematic review and a meta-analysis.

Systematic review vs Meta-analysis (5 mins)



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=1720#oembed-1>

Systematic review vs Meta-analysis (5 mins) by Research Masterminds (YouTube)

Some brief information about other types of reviews, such as scoping reviews, rapid reviews, and meta-analyses is available from the library guide: Systematic Reviews.

Benefits of a systematic review

A systematic review has several advantages, including:

- Being considered the highest level in the hierarchy of literature evidence.
- Provides a definitive answer to a question, commonly about therapy, prevention, and causes of disease or harm.
- The methods used to find and select studies reduce bias and are more likely to produce reliable and accurate conclusions.
- Summarises findings from multiple studies, reducing bias when drawing conclusions and making the findings more reliable.
- Findings might be applied to everyday practice or to inform policy.
- Identifies knowledge gaps that call for more research.



"File:Evidence-based Medicine (EBM).png" by Tingjoh is licensed under CC BY-SA 4.0

Benefits for HDR candidates

Conducting a systematic review provides several benefits:

- it fulfills the request made by your supervisor to conduct the systematic review
- it contributes to your thesis
- the paper can be published
- it contributes to your research knowledge of the topic.

Test your knowledge



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=1720#h5p-14>

DEVELOPING THE RESEARCH QUESTION

A systematic review is an in-depth attempt to answer a specific, focused question in a methodical way.

A clearly defined research question should accurately and succinctly sum up the review's line of inquiry.



Photo by [Tingey Injury Law Firm](#) on Unsplash

In developing the research question ensure that it is not just a topic, but a properly formulated question that is answerable.

Consider whether your question will focus on diagnosis, intervention, prognosis, or etiology? Is there a study design (e.g. Randomised Controlled Trials) that would provide the best answer?

A good question will combine several concepts. Identifying the relevant concepts is crucial to the successful development and execution of your systematic review. Your research question should provide you with a checklist of the main concepts to be included in your search strategy.

If appropriate, use a framework to help you develop your

research question. A framework will assist in identifying the important concepts in your question.

One technique often used to help formulate a clinical research question is the PICO model.

P = Population / Patient / Problem

I = Intervention / Indicator

C = Comparison / Control

O = Outcome

There are other frameworks such as SPICE, SPIDER, and ECLIPSE. More information on these frameworks is available from the library guide: Systematic Reviews.

Activity

If you were undertaking a systematic review and researching the effectiveness of acupuncture for

treating allergic rhinitis what would your research question be?



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<https://rmit.pressbooks.pub/researchwritingmodules/?p=1733#h5p-17>

Think about one of your research questions. How might you adjust the question by applying the use of the PICO framework?

Best Practice Tip

Prior to commencing the systematic review, first, determine if a similar review has been recently done.

You could do this by searching relevant databases or the PROSPERO register of systematic reviews.

WRITING THE PROTOCOL

A protocol should be prepared *before* a review is started and used as a guide to carry out the review.



Photo by [Laura Chouette](#) on Unsplash

The aim of the protocol is to minimise bias by having pre-defined eligibility criteria of what *will*, and *will not*, be included in the review.

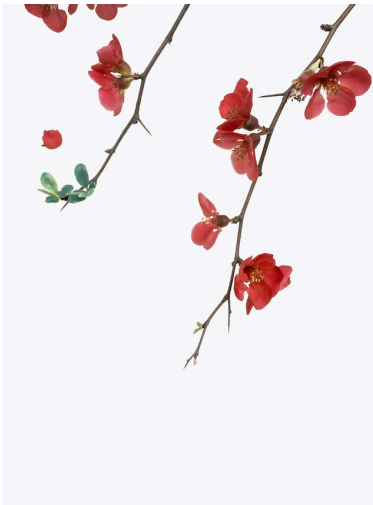
The research protocol is a planning document that will:

- describe the rationale for the review
- set out the review objectives
- detail the sources and search strategy used to locate studies
- detail how studies will be selected based on the defined eligibility criteria for the

- inclusion/exclusion of studies
- detail how the studies will be critically analysed
- provide the basis of how the findings will be reported.

The protocol is developed in conjunction with determining search terms.

A protocol promotes research integrity, accountability, and transparency of the completed review.



Best Practice Tip

It is recommended that you use a standard such as the 27 item PRISMA checklist to develop your protocol. This document will then serve well as a guide to what should

Photo by [Han Chenxu](#) on [Unsplash](#)

be included when the findings of the systematic review are reported.

What is PRISMA?

PRISMA is the Preferred Reporting Items for Systematic reviews and Meta-Analyses. PRISMA is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses.

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=1736#h5p-15>

PLANNING YOUR SEARCH STRATEGY

Planning your search is critical to the success of the systematic review with the following iterative steps included:

1. Where to search, locating the appropriate sources to search, which will work, which won't.
2. How best to effectively search, to develop the search terms and how they will be combined.
3. Testing that your search strategy will yield the anticipated results to answer the research question.

Sources to search

The search for literature for a systematic review should be rigorous and comprehensive to find ALL information available on a particular topic. It is therefore important to widely and thoroughly search published and unpublished research.



Photo by Jan Antonin Kolar on
Unsplash

There are several types of sources that you can search, including; databases, grey literature, trials, and reference lists. You can also try hand-searching.

For HDR Candidates

As an HDR candidate, you will most likely only use databases for your systematic review, although

confirm with your supervisor their expectations on which sources to search.

Databases – It is important to search across a range of databases as no one database covers all the related literature. It is not acceptable to search just one database. The decision regarding which databases to search depends on the topic of the review. The database searches need to be comprehensive and reproducible.

Grey Literature – This is not controlled by commercial publishers but rather is produced by organisations, governments, and industry. Grey literature is less likely to exhibit publication bias and so can provide balance.

Trials – Many clinical trials are unpublished, so when appropriate it is important to include unpublished and ongoing studies to minimise bias.

Hand searching – Not all trial reports are included in bibliographic databases, and trials may not be easily identified in database search results when accessing the titles and abstracts. Hand searching is a manual page-by-page examination of relevant journals and conference proceedings in order to identify published trials.

Reference lists – It can be fruitful to search the reference lists of relevant systematic reviews and meta-analyses, as well as

other key identified studies. Using this search method would be done in the preliminary stages to help determine that search results did contain these papers.

Planning your literature search

The planning phase of developing, testing, and revising your search queries is crucial to the success of the systematic review.

Do you have a set of relevant papers already that you want to include in your systematic review?

It is useful to build a ‘sample set’ of relevant references before you develop your search strategy. The ‘sample set’ may include:



Photo by [Andrew Neel](#) on [Unsplash](#)

- key papers recommended by your supervisor
- references used in similar systematic reviews

The ‘sample set’ of references will enable you to:

- help identify relevant search terms
- test that your search strategy will retrieve these references

(and subsequently other relevant references on your research topic)

Developing, testing, and revising your search strategy

The search strategy needs to include a detailed list of **search terms** for each concept to ensure all relevant studies are captured for the review. Search terms will be made up of keywords or phrases, as well as database subject headings. Each database uses a different criterion to classify articles, so the subject headings will differ between them.

When using multiple databases, you are likely to encounter a large volume of resources. When planning your searches, you should continually adjust search terms and/or selection criteria in order to make sure you have a comprehensive body of references.

It is recommended that you **test your search terms** to determine if all the subject headings and words/phrases will return useful results. Test your search strategy in a key database. Does it retrieve any papers from your ‘sample set’ that are contained in that database? Are the results of the search relevant to your topic? What proportion are irrelevant? Identify any terms that are retrieving large numbers of irrelevant papers.

Documenting your search

It is essential that you thoroughly document your search process in enough detail to ensure that it can be reported correctly in the review. An Excel spreadsheet is one tool that you could consider using to document your searching.

For each database search, you should record:

- the date the search was run
- the name of the database
- the name of the database provider (e.g. ProQuest or EBSCO)
- your search strategy – include the keywords you used and how these were combined in the search
- any filters or limitations used, such as years, language, etc.
- the number of studies identified

PLANNING YOUR SEARCH TERMS

Define your research question

If you were researching the effectiveness of acupuncture for treating allergic rhinitis, you might begin with a research question that looks like the following.

Example research question

How effective is acupuncture in treating allergic rhinitis?

This question could be improved by utilising the PICO framework to look like this:

Improved example research question

In patients with allergic rhinitis is acupuncture compared with pharmacological treatment more effective in decreasing pain symptoms?

P (problem) = In patients with allergic rhinitis

I (intervention) = is acupuncture

C (comparator) = compared with pharmacological treatment

O (outcome) = more effective in decreasing pain symptoms?

Compiling a list of search terms

Let's look at two of the concepts from the research question:

1. **allergic rhinitis**
2. **acupuncture**

When you start compiling a list of possible search terms you need to think of your own keywords, as well as the thesaurus (i.e. subject) terms used by *each* of the databases you will be searching.

The following table shows some keywords and thesaurus terms for allergic rhinitis and acupuncture. The thesaurus terms are MeSH (Medical Subject Headings) as used by the PubMed database. Thesaurus terms for other databases would be added to the list.

Database search terms

Concepts	Allergic rhinitis	Acupuncture
Keywords		plum blossom ear acupuncture ear acupressure auricular therapy moxa laser acupuncture seven star needle electro-acupuncture electro acupuncture TENS transcutaneous nerve stimulation transcutaneous electrical nerve stimulation electro-stimulation electro stimulation pharmacopuncture point injection
	allergic rhinitis hayfever hay fever pollinosis rose cold rose fever perennial hayfever seasonal hayfever	
MeSH (Medical Subject Headings)	rhinitis, allergic rhinitis, allergic, seasonal rhinitis, allergic, perennial respiratory hypersensitivity	acupuncture meridians electroacupuncture moxibustion auriculotherapy acupressure acupuncture, ear acupuncture therapy acupuncture analgesia acupuncture points transcutaneous electric nerve stimulation

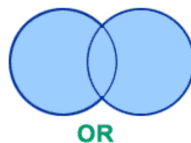
COMBINING YOUR SEARCH TERMS

Combine with Boolean operators

To combine the different search terms the boolean operators of OR and AND need to be used.

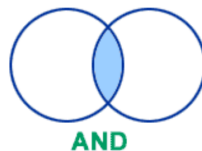
OR operator broadens your search

Use the boolean operator **OR** to combine keywords related to a single concept. This **broadens your search**, increasing the number of results. This means that each search term listed for allergic rhinitis would be combined with an OR, and similarly for acupuncture.



AND operator narrows your search

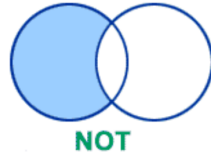
Use the boolean operator **AND** to combine keywords related to different concepts. This **narrows your search**, decreasing the number of results. The search results for allergic



rhinitis would be combined with an AND with the search results for acupuncture.

NOT operator narrows your search

Use the boolean operator **NOT** operator to exclude a term. This is useful if you don't want to find any records that contain a particular term. Use it with care to avoid excluding relevant articles that briefly mention the second term. Searching for allergic rhinitis NOT children will exclude all records that mention children as well as allergic rhinitis and children together.



Test your knowledge



An interactive H5P element has been excluded from this version of the text. You

can view it online here:

*[https://rmit.pressbooks.pub/
researchwritingmodules/?p=2195#h5p-19](https://rmit.pressbooks.pub/researchwritingmodules/?p=2195#h5p-19)*

The search query

The search features of library databases make constructing your search query easier, especially when using advanced search options. For example, the Boolean operators can usually be selected from a drop-down menu. Some databases allow you to enter the search terms on separate lines of an Advanced or Basic search form. Concepts can be entered on separate lines that are then combined using the OR and AND operators.

Best Practice Tip

If there are many synonyms and subject headings for a single concept, you may find it helpful to break this concept into a couple of separate searches that you can then combine using

OR, as in the example below. The first concept is **allergic rhinitis** and the second concept is **acupuncture**.

Search query

Search line	Terms combined	Notes
1	“Allergic rhinitis” OR hayfever OR “hay fever” OR pollinosis OR “rose fever” OR “perennial hayfever” OR “seasonal hayfever”	Concept 1 keyterms
2	“Rhinitis, Allergic”[Mesh] OR “Rhinitis, Allergic, Seasonal”[Mesh] OR “Rhinitis, Allergic, Perennial”[Mesh] OR “Respiratory Hypersensitivity”[Mesh]	Concept 1 MeSH headings
3	2 OR 3	All Concept 1
4	“plum blossom” OR “ear acupuncture” OR “ear acupressure” OR auricular therapy OR moxa OR “laser acupuncture” OR “seven star needle” OR “electro-acupuncture” OR “electro acupuncture” OR TENS OR “transcutaneous electrical nerve stimulation” OR “transcutaneous nerve stimulation” OR “electrostimulation” OR pharmoacupuncture OR “point injection”	Concept 2 keyterms
5	“Acupuncture”[Mesh] OR “Meridians”[Mesh] OR “Electroacupuncture”[Mesh] OR “Moxibustion”[Mesh] OR “Auriculotherapy”[Mesh] OR “Acupressure”[Mesh] OR “Acupuncture, Ear”[Mesh] OR “Acupuncture Therapy”[Mesh] OR “Acupuncture, Analgesia”[Mesh] OR “Acupuncture Points”[Mesh] OR “Transcutaneous Electric Nerve Stimulation”[Mesh]	Concept 2 MeSH headings
6	4 OR 5	All Concept 2

Search line	Terms combined	Notes
7	3 AND 6	Combined concepts

Our search query using the PubMed database and the listed keywords and MeSH terms for allergic rhinitis, and similarly for acupuncture, would look like the following.

Example search query

“Allergic rhinitis” OR hayfever OR “hay fever” OR
pollinosis OR “rose fever” OR “perennial hayfever”
OR “seasonal hayfever” OR “Rhinitis,
Alllergic”[Mesh] OR “Rhinitis, Allergic,
Seasonal”[Mesh] OR “Rhinitis, Allergic,
Perennial”[Mesh] OR “Respiratory
Hypersensitivity”[Mesh] **AND** “plum blossom” OR
“ear acupuncture” OR “ear acupressure” OR auricular
therapy OR moxa OR “laser acupuncture” OR “seven
star needle” OR “electro-acupuncture” OR “electro
acupuncture” OR TENS OR “transcutaneous

electrical nerve stimulation" OR "transcutaneous
 nerve stimulation" OR "electrostimulation" OR
 pharminoacupuncture OR "point injection" OR
 "Acupuncture"[Mesh] OR "Meridians"[Mesh] OR
 Electroacupuncture"[Mesh] OR "Moxibustion"[Mesh]
 OR "Auriculotherapy"[Mesh] OR
 "Acupressure"[Mesh] OR "Acupuncture, Ear"[Mesh]
 OR "Acupuncture Therapy"[Mesh] OR "Acupuncture,
 Analgesia"[Mesh] OR "Acupuncture Points"[Mesh]
 OR 'Transcutaneous Electric Nerve
 Stimulation"[Mesh]

The library guide [Starting your literature review](#) has a page that addresses planning your search. It covers identifying the main aspects/concepts of your topic, compiling a list of search terms, utilising different search techniques, and the search query.

ANALYSING SEARCH RESULTS

Once the search results are obtained the analysis phase of the systematic review commences where each paper must be judged on validity and quality. The decisions on which papers to include and exclude are based on the eligibility criteria specified in the protocol.

Best Practice Tip

It is recommended that you use the PRISMA flow diagram to record the number of search results from each database and the changes to those numbers as the screening process is conducted. This diagram should be included in the reported findings of the systematic review.

What is PRISMA?

PRISMA is the Preferred Reporting Items for Systematic reviews and Meta-Analyses. PRISMA is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses.

Screening papers

The screening process of a systematic review will usually involve more than one reviewer, with the reviewers assessing each paper against the eligibility criteria for inclusion or exclusion from the review as defined in the protocol.

Each reviewer will need to provide a rationale for how the papers match the eligibility criteria or have some value to the systematic review. This is done separately by each reviewer to ensure minimal bias. The results are then compared.

The screening is done incrementally in two phases. An agreement must be reached by all reviewers, at each phase, on which papers are to be included and excluded.

Phase 1 – **Titles and abstracts** of each paper are assessed against the eligibility criteria.

Phase 2 – The process is repeated where the **full text** of each of the remaining papers are assessed against the eligibility criteria.



Photo by Nong V on Unsplash

The remaining papers are those that will contribute to the systematic review.

Critical appraisal

The critical appraisal process examines the validity and applicability of the studies. It is important to critically evaluate the literature to:

- assess the benefits and strengths of the research against flaws and weaknesses
- decide whether studies have been undertaken in a way that makes their findings reliable
- make sense of the results
- know what these results mean in the context of the clinical decision being made
- assess the usefulness of the evidence for clinical decisions.

There are several **checklists** available to assist in the process of determining the quality of the studies. A checklist should be chosen to assess sources of bias that are likely to affect results in relation to the research question.

Take a look at the following sites for examples of critical appraisal tools that can be used to assess the quality, validity, and bias of papers.

- CASP (Critical Appraisal Skills Programme) Checklists
- Joanna Briggs Critical Appraisal Tools
- Risk of bias assessment tools from the University of Bristol

SYNTHESISING THE DATA

Synthesis is a stage in the systematic review process where extracted data, that is the findings of individual studies, are combined and evaluated.

The general purpose of extracting and synthesising data is to show the outcomes and effects of various studies, and to identify issues with methodology and quality. This means that your synthesis might reveal several elements, including:

- overall level of evidence
- the degree of consistency in the findings
- what the positive effects of a drug or treatment are, and what these effects are based on
- how many studies found a relationship or association



Photo by [Pawel Czerwinski](#) on
Unsplash

between two components, e.g. the impact of disability-assistance animals on the psychological health of workplaces

There are two commonly accepted methods of synthesis in systematic reviews:

1. Qualitative data synthesis
2. Quantitative data synthesis (i.e. meta-analysis)

The way the data is extracted from your studies, then synthesised and presented, depends on the type of data being handled.

Qualitative data synthesis

In a qualitative systematic review, data can be presented in a number of different ways. A typical procedure in the health sciences is **thematic analysis**.

Thematic synthesis has three stages:

1. the coding of text ‘line-by-line’
2. the development of ‘descriptive themes’
3. and the generation of ‘analytical themes’

If you have qualitative information, some of the more common tools used to summarise data include:

- textual descriptions, i.e. written words
- thematic or content analysis

Example qualitative systematic review

A good example of how to conduct a thematic analysis in a systematic review is the following journal article on cancer patients. In it, the authors go through the process of:

1. identifying and coding information about the selected studies' methodologies and findings on patient care
2. organising these codes into subheadings and descriptive categories
3. developing these categories into analytical themes

What Facilitates “Patient Empowerment” in Cancer Patients During Follow-Up: A Qualitative Systematic Review of the Literature

Quantitative data synthesis

In a quantitative systematic review, data is presented statistically. Typically, this is referred to as a **meta-analysis**.

The usual method is to combine and evaluate data from multiple studies. This is normally done in order to draw conclusions about outcomes, effects, shortcomings of studies and/or applicability of findings.

Remember, the data you synthesise should relate to your research question and protocol (plan). In the case of quantitative analysis, the data extracted and synthesised will relate to whatever method was used to generate the research question (e.g. PICO method), and whatever quality appraisals were undertaken in the analysis stage.

If you have quantitative information, some of the more common tools used to summarise data include:

- grouping of similar data, i.e. presenting the results in tables
- charts, e.g. pie-charts
- graphical displays, i.e. forest plots

Example of a quantitative

systematic review

A quantitative systematic review is a combination of qualitative and quantitative, usually referred to as a meta-analysis.

Effectiveness of Acupuncturing at the Sphenopalatine Ganglion Acupoint Alone for Treatment of Allergic Rhinitis: A Systematic Review and Meta-Analysis

About meta-analyses

A systematic review may sometimes include a meta-analysis, although it is not a requirement of a systematic review. Whereas, a meta-analysis also includes a systematic review.

A meta-analysis is a

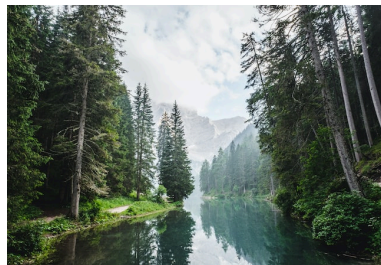


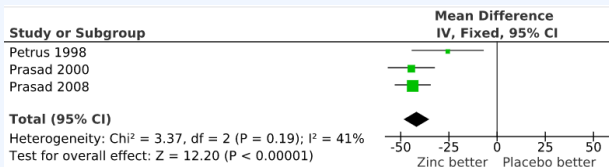
Photo by [Luca Bravo](#) on Unsplash

statistical analysis that combines data from previous studies to calculate an overall result.

One way of accurately representing all the data is in the form of a **forest plot**. A forest plot is a way of combining the results of multiple studies in order to show point estimates arising from different studies of the same condition or treatment.

It is comprised of a graphical representation and often also a table. The graphical display shows the mean value for each study and often with a confidence interval (the horizontal bars). Each mean is plotted relative to the vertical line of no difference.

The following is an example of the graphical representation of a forest plot.



“File:The effect of zinc acetate lozenges on the duration of the common cold.svg” by Harri Hemilä is licensed under CC BY 3.0

Watch the following short video where a social health example is used to explain how to construct a forest plot graphic.

Forest Plots: Understanding a Meta-Analysis in 5 Minutes or Less (5:38 mins)



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=1746#oembed-1>

Forest Plots – Understanding a Meta-Analysis in 5 Minutes or Less (5:38 min)
by The NCCMT (YouTube)

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=1746#h5p-16>

REPORTING THE FINDINGS

While the writing process for a systematic review is generally like writing any other kind of review, there are several aspects to note.



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In writing the systematic review you should provide an **answer to the research question**.

Careful **documentation of the methodology** is important as it should outline the search process and the selection process. A reader should understand why sources were chosen, how they were assessed, and how conclusions were reached.

The **structure** of the systematic review will differ from the traditional (or narrative) literature review as it should **reflect the stages outlined in the protocol**. Refer to the 27 item PRISMA checklist to see what should be addressed in the protocol.

The value of a systematic review is the **critical reflection and interpretation of the findings**.

Reporting the findings of the systematic review will differ slightly if it is to be presented as part of a thesis, or as a manuscript for publication.

Examples

The following examples are available from the RMIT Research Repository.

Dissertation / Thesis

- Ear-acupressure for allergic rhinitis

Articles

- Ear-acupressure for allergic rhinitis: A systematic review
- Acupressure for respiratory allergic diseases: A systematic review of randomised controlled trials

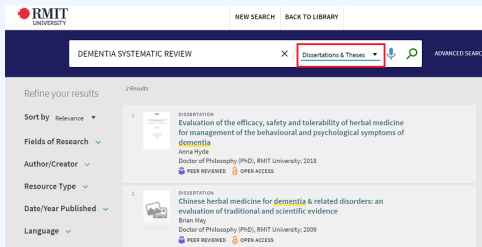
Activity

To see how a systematic review is written check out examples of published papers and/or completed theses.

1. Find a systematic review by searching a database and examine how the review has been written. For example, search the PubMed database on your topic and filter results by 'article type' selecting 'systematic reviews.'
2. Find a systematic review paper by searching with the words 'systematic review' in the RMIT Research Repository. Consider adding an additional topic word.
3. HDR candidates may like to ask their supervisor if they can recommend a completed thesis that includes a systematic review.

The following image shows how to search the Research

Repository for a thesis with a systematic review if an author's name is unknown. In the search box include a topic along with the words 'systematic review' and select 'Dissertations & Theses' from the drop-down menu.



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)

EXPLORE FURTHER

Congratulations on completing this module. We have explored the benefits of conducting a systematic review and looked at the differences between a traditional literature review and a systematic review.



Photo by [Erwan Hesry](#) on [Unsplash](#)

Various aspects of a systematic review were briefly examined, including; the research question and PICO; developing a protocol and PRISMA; resources and strategies for searching the literature; eligibility criteria for screening results; critical appraisal tools to assess the quality of studies; synthesizing data; and writing the review.

Further assistance

Individual assistance – For more assistance in conducting a systematic review contact the library by submitting a Request Research Assistance form.

Library guide – Additional information is available from the online library guide [Systematic Reviews](#).

Cochrane handbook – The Cochrane Handbook for Systematic Reviews of Interventions details systematic reviews more thoroughly, including meta-analysis. It is the official guide and describes in detail the process of preparing and maintaining Cochrane systematic reviews on the effects of healthcare interventions. The *Handbook* includes guidance on the standard methods applicable to every review (planning a review, searching and selecting studies, data collection, risk of bias assessment, statistical analysis, GRADE and interpreting results), as well as more specialised topics (non-randomized studies, adverse effects, complex interventions, equity, economics, patient-reported outcomes, individual patient data, prospective meta-analysis, and qualitative research).

Learn more

A short video Introducing systematic reviews (14 mins) is available (Microsoft Stream, login required).

You may also be interested in viewing the following webinar Systematic Reviews (60 mins) recorded from the Library's PhD Up! program (now called Research Plus). The webinar content complements this online module.

Feedback

Your comments and suggestions on how we can improve this

module will be appreciated. Please complete our feedback form.

What's next?

You have completed this module on Conducting a Systematic Review. You can return to the Research and Writing Hub to choose another module to complete.

DEVELOPING AN EFFECTIVE RESEARCH PROPOSAL

Welcome to this RMIT University Library online module which provides you with key information, strategies and resources for developing an effective research proposal.

The module is aimed at graduate researchers (PhD or Masters by Research) who are seeking to

understand why and how to write their research proposal.

The course materials include information, videos and learning activities which explore key features of developing effective research proposals.

We hope you find this module useful and stimulating.



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Learning objectives

On successful completion of the module, you will be able to:

1. determine the purpose and importance of the research proposal
2. create effective research questions
3. identify the key components which structure a research proposal, and
4. familiarise yourself with writing the “what”, “why” and “how” sections of your proposal.

This module should take you about 30 minutes to complete.

Work through each chapter using the navigation footer (i.e. previous/next) or use the contents page to select a particular section.

Please help us improve this module by taking a moment to complete the feedback form provided at the end.

WHAT IS A RESEARCH PROPOSAL?

Overview

A research proposal is a type of text which maps out a proposed central research problem or question and a suggested approach to its investigation.

In many universities, including RMIT, the research proposal is a formal requirement. It is central to achieving your first milestone: your Confirmation of Candidature. The research proposal is useful for both you and the University: it gives you the opportunity to get valuable feedback about your intended research aims, objectives and design. It also confirms that your proposed research is worth doing, which puts you on track for a successful candidature supported by your School and the University.

Although there may be specific School or disciplinary requirements that you need to be aware of, all research proposals address the following central themes:

- **what** you propose to research
- **why** the topic needs to be researched
- **how** you plan to research it.

Purpose and audience

Before venturing into writing a research proposal, it is important to think about the **purpose** and **audience** of this type of text. Spend a moment or two to reflect on what these might be.

Reflection

What do you think is the purpose of your research proposal and who is your audience?

Purpose

The purpose of your research proposal is:

1. To allow experienced researchers (your supervisors and their peers) to assess whether

- the research question or problem is viable (that is, answers or solutions are possible)
- the research is worth doing in terms of its contribution to the field of study and benefits to stakeholders
- the scope is appropriate to the degree (Masters or PhD)
- you've understood the relevant key literature and identified the gap for your research
- you've chosen an appropriate methodological approach.

2. To help you clarify and focus on what you want to do, why you want to do it, and how you'll do it. The research proposal helps you position yourself as a researcher in your field. It will also allow you to:

- systematically think through your proposed research, argue for its significance and identify the scope
- show a critical understanding of the scholarly field

around your proposed research

- show the gap in the literature that your research will address
- justify your proposed research design
- identify all tasks that need to be done through a realistic timetable
- anticipate potential problems
- hone organisational skills that you will need for your research
- become familiar with relevant search engines and databases
- develop skills in research writing.

Audience

The main audience for your research proposal is your reviewers. Universities usually assign a panel of reviewers to which you need to submit your research proposal. Often this is within the first year of study for PhD candidates, and within the first six months for Masters by Research candidates.



Photo by [Ahmed Galal](#) on Unsplash

Your reviewers may have a strong disciplinary understanding of the area of your proposed research, but depending on your specialisation, they may not. It is therefore important to create a clear context, rationale and framework for your proposed research. Limit jargon and specialist terminology so that non-specialists can comprehend it. You need to convince the reviewers that your proposed research is worth doing and that you will be able to effectively 'interrogate' your research questions or address the research problems through your chosen research design.

Your review panel will expect you to demonstrate:

- a clearly defined and feasible research project
- a clearly explained rationale for your research
- evidence that your research will make an original contribution through a critical review of the literature
- written skills appropriate to graduate research study.

DEVELOPING A RESEARCH QUESTION

Your research questions are perhaps the most important part of your study. They guide your choice of methodology and underpin each chapter in your thesis or dissertation. It's worth investing time to develop robust questions that will guide your research.

Identifying an effective research question

While the criteria for an effective research question vary considerably, generally, a research question should be:

- **focused** – its scope should be adequately narrow to allow you to carry out your research within the available timeframe and using available resources.

- **analytical (as opposed to descriptive)**
 - your research question needs to display enough complexity so that the answers to it cannot be easily obtained. For example, it cannot be answered through a simple internet search or fact check. An effective research question would require answers which are subject to interpretation, analysis and synthesis.
- **effectively expressed** – the research question needs to use clear, specific and concise language so that it is accessible to the reader.

Activity



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=233#h5p-3>

Strategies for developing effective research questions

To develop effective research questions, you may like to try one of the following two key strategies:

I) Convert your topic into one or more research questions by:

1. breaking down your topic into its different features
2. choosing a feature that interests you to narrow down your topic scope
3. brainstorming and reading literature around this feature to focus it further
4. convert this focused feature into question form.



Photo by [Tingey Injury Law Firm](#) on Unsplash

II) Formulate a problem statement and then convert it into question form. Use the following template as a guide to writing your problem statement:

1. I am examining ...
2. Because ...
3. It matters because...

The following presentation shows how to use the above discussed strategies for developing robust research questions.

Work through each section of the webinar. Feel free to pause the video at suitable points and complete the included activities.

Presentation on developing research questions (12:54 min)



One or more interactive elements has been excluded from this version of the text. You can view them online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=233#oembed-1>

Developing your research questions (12:54 min) by RMIT University (YouTube)

Further resources

For more information on research questions, consult the following resources:

- How Research Questions Can Make or Break Your Project (8:36 min) by Professor James Arvanitakis (YouTube)
- Research Questions Hypothesis and Variables: Connecting the Dots (7:55 min) by Associate Professor Ron Wallace (YouTube)

STRUCTURING KEY SECTIONS

All research proposals need to cover the following three main areas:

- **what** you propose to research
- **why** the topic needs to be researched
- **how** you plan to research it.

How these three areas translate into a structure or into sections in a research proposal varies significantly.

There isn't usually a single set structure for all research proposals, so you'll need to check your School guidelines. There is also substantial disparity in the length required, with some Schools asking for 2 – 5 pages, some 8 – 10 pages, and others considerably longer.

The specific sections required in research proposals can also vary between Schools.

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=880#h5p-1>

To help you work out how you went with the above activity, the table below lists sections that are **always included** and those that are **often included** in a research proposal.

Always included	Often included
• Title • Background to the study • Literature review • Research questions, problems and/or hypotheses • Rationale for the research • Scope of the research • Research methodology • Milestones or timeline • Research significance/contributions • Reference list	• Abstract • Table of Contents • Introduction • Definitions • Statement of the problem • Research aims and/or objectives • Epistemological stance and/or theoretical framework • Particular needs (e.g. resources) • Expected preliminary outcomes • Ethics approval/evidence of application

For some general RMIT discipline-based guidelines, consult RMIT's discipline based guidelines to writing research proposals (DOCX, 1 page).

WRITING YOUR RESEARCH PROPOSAL

Writing the 'what' of your proposed research

The purpose of this part of your research proposal is to generally describe what your research is about.



[Photo by Kaitlyn Baker on Unsplash](#)

The 'what' part establishes how your research is situated within your discipline or field. It provides fundamental information, such as:

- the context for the research, which can be:
 - the key ideas, theories and concepts
 - the major issues and debates
 - the key players and seminal texts or key artists, and
 - the questions that have been asked around this topic.

- your research questions, problems or hypotheses
- the scope of the proposed research (i.e. what you will and what you won't do).

The 'what' part of your research proposal may also include:

- the aims and/or objectives of your research
- an introduction to the theoretical framework within which your research sits
- a statement of the problem grounded in the context or theoretical framework and a resulting argument for your research to be conducted
- the timeliness of your proposed research (i.e. why should it be done right now?)
- definitions if needed.

Writing the 'why' of your proposed research

A key requirement of your research proposal is to justify that your research is worth doing. Your review panel will be looking for a succinct and convincing argument about what sets your proposed research apart from others, and why *not* doing this research leaves an important problem unaddressed.

Ways of justifying your research include showing that your

project will make a significant and substantial contribution in terms of:

1. how it fits within an existing body of scholarship/
literature/practice
2. how it builds on and adds to this body of knowledge
3. what the value of your research is and for whom (e.g. a particular community, industry, etc.).

Reflection

Think about how you might go about justifying your research.

- What key literature is your proposed research situated within?
- How will your research build on this?
- Who will benefit from your research and how?

Critical engagement with the literature is crucial in order to

justify your research. You must demonstrate that you understand:

- the main concepts and themes, underlying principles, and established theories related to your research
- areas of controversy and contention
- the key scholars and seminal research related to your topic.

In some disciplines, discussion of the above points is located in the ‘what’ rather than the ‘why’ section of your research proposal. In fact, you will almost certainly cover some of these points when you’re describing and contextualising your research. Often this is done in a general way in the ‘what’ section and in a more critical, in-depth way in the ‘why’ section. A rule of thumb is to ask yourself whether this information is **contextualising** or **justifying** your research.

Writing the ‘how’ of your proposed research

This part of your research proposal involves describing how you plan to find answers to your research questions or resolve the research problems. In other words, it entails describing the design of your research.

A difficulty you may have in this section is in providing the

right amount of information – not too little nor too much. You need to give as much information as is needed to argue to the review panel that the research is do-able and to justify the components of your research design.

Reflection

Which of these questions will you need to answer in your research design (i.e. the 'how' section of your research proposal)?

- What is your chosen research design and rationale?
- What theories, concepts or models inform your research design?
- What are the step-by-step methods or process used?
- What constitutes your creative practice?
- How will you engage with your creative practice (e.g. reflection, testing, theorising)?

- What type of data will be collected?
- What are your sources of data?
- Where and how will the data be collected?
- How will the data be analysed?
- What are the strengths and limitations of your methodology?
- What resources are required (equipment, other)?
- How reliable and valid are your methods?
- What ethical issues relate to your research methods, and how will you address these?
- Can you complete your research within the official timeframe (demonstrated on a Gantt chart)?

Learn More

To learn more about writing a research proposal, watch this webinar from the Library.

RMIT webinar on writing a research proposal (23:59)





One or more interactive elements has been excluded from this version of the text. You can view them online here:
<https://rmit.pressbooks.pub/researchwritingmodules/?p=239#oembed-1>

Writing a research proposal (23:59 min) by RMIT University Library (YouTube).

Further resources

The Manchester Academic Phrase Bank provides a list of ‘starter phrases’ which can be very helpful for writing different sections of your research proposal. The Bank’s Introducing Work and Referring to Sources sections may be particularly useful for the ‘what’ and ‘why’ parts of the proposal while Describing Sources would be useful for the ‘how’ part.

For some RMIT discipline-based guidelines for research proposals illustrating the ‘what’, ‘why’, and ‘how’ sections, consult: Discipline- based guidelines (DOCX, 1 page).

EXPLORE FURTHER

Congratulations on completing this module. We have explored the nature, purpose, and key component structures of research proposals. We have also discussed some strategies for writing effective research questions as well as the ‘what’, ‘why’ and ‘how’ sections of a proposal.



Photo by Jason Leung on Unsplash

Further assistance

For more assistance in planning or writing your research proposal, consult the following resources:

- How Research Questions Can Make or Break Your Project (8:36 min) by Professor James Arvanitakis (YouTube)
- Research Questions Hypothesis and Variables: Connecting the Dots (7:55 min) by Associate Professor Ron Wallace (YouTube)

- The Manchester Academic Phrase Bank

You can also contact the Library by submitting a research request form.

Feedback

Your comments and suggestions on how we can improve this module will be appreciated. Please complete our feedback form.

What's next?

You have completed this module on the Research Proposal. To choose another module to complete, you can return to the Research and Writing Hub.

SETTING UP YOUR RESEARCHER PROFILE

Welcome to this online module on how to set up your researcher profile. Establishing a researcher profile is very useful for promoting and increasing the visibility of your research publications and other activities.



Photo by [Ricardo Gomez Angel](#) on Unsplash

The module provides an overview of commonly used researcher profiles and how to establish and maintain them.

Learning objectives

On successful completion of the module, you will:

- be familiar with the most commonly used researcher profiles
- know the benefits of researcher profiles
- be able to identify which profiles will be most relevant to you
- know how to create and maintain a researcher profile, as well as connect it to other online profiles.

The course materials include information, videos, and learning activities that explore key aspects of establishing your researcher profile.

This module should take you about 30 minutes to complete.

Work through each section using the navigation footer (i.e. previous/next), or use the contents menu to select a particular section.

Upon completion, please help us improve this module by

taking a moment to complete the feedback form provided at the end.

You may also be interested in viewing the additional resources about setting up a researcher profile in the Explore Further chapter at the end of this module. Provided is a webinar recorded from the Library's 'PhD Up!' program (now called Research Plus), along with the library guide on Researcher Profiles and ORCID iDs.

ABOUT RESEARCHER PROFILES

What is a researcher profile?

A researcher profile is the publicly accessible profile of your professional academic works and achievements. A researcher profile enables information about you, your work and career to be

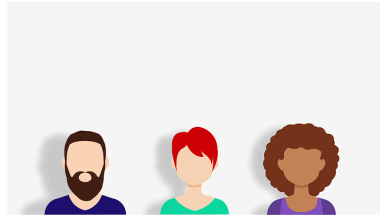


Image by Coffee Bean from Pixabay

visible and accessible, encouraging collaboration, and providing measures of your impact and engagement. A researcher profile will bring together your research outputs on a single platform, making your work easier to find.

An **identifier** is a unique number (e.g. 16 digit ORCID iD), or persistent URL. Whilst a **profile** is the body of information that is associated with an identifier.

Benefits of a researcher profile

Each researcher profile has different strengths, together they can:

- increase the visibility of your research
- improve your chance of being cited
- improve your citation metrics
- ensure correct attribution, reducing the number of name variations
- connect you to new collaborators / funders
- increase your employment opportunities
- increase your standing within your field of study beyond RMIT University
- link your work from different sites to one place.

Types of profiles

Researcher profiles are an essential component of an academic career and there are a few types that can be created and interlinked.

This module will explore the common researcher profiles:

- ORCID iD
- Scopus Author ID
- Web of Science Researcher ID
- Google Scholar Profile

ORCID: A MUST HAVE PROFILE

What is ORCID?

The ORCID iD, formed from the acronym **O**pen **R**esearcher and **C**ontributor **I**Dentifier, is a unique identifier that distinguishes you from every other researcher. Unlike other researcher profiles, ORCID is controlled by

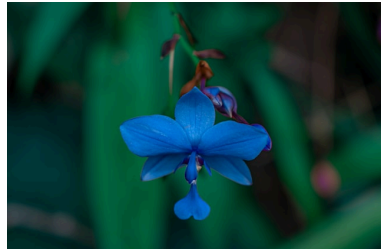


Photo by Ramith Bhasuka on
Unsplash

you and can be a comprehensive reflection of your research.

Use ORCID to display your:

- affiliation and employment
- research interests and projects
- invited positions, awards and prizes
- membership of academic or professional societies
- grant funding
- research outputs such as journal articles, books and book

chapters, conference papers, as well as other non-traditional research outputs (NTROs) such as creative works.

Example ORCID profile

Meredith Tharapos' ORCID profile is an example of how to add content to different fields.

ORCID iDs can also be used to connect with:

- publishers – ORCID iDs are often requested when submitting manuscripts and conference papers
- other online profiles such as your RMIT profile, LinkedIn, personal website, or database profile
- research funding systems such as the ARC's Research Management System (RMS).

Watch this video to discover more about ORCID iDs and the benefit of creating one for yourself.

What is ORCID? (4:17 mins)



One or more interactive elements has been excluded from this version of the text. You

can view them online here:

*[https://rmit.pressbooks.pub/
researchwritingmodules/?p=2323#oembed-1](https://rmit.pressbooks.pub/researchwritingmodules/?p=2323#oembed-1)*

What is ORCID? (4:17 mins) from ORCID on Vimeo.

Reflection

If you already have an ORCID profile how might you improve it?

What other information might you add to make your profile more comprehensive?

For HDR Candidates

Find an ORCID profile for a researcher at RMIT, maybe your supervisor's profile.

How could they improve on their profile?

What other information could they add to make their profile more comprehensive?

CHOOSING YOUR RESEARCHER PROFILES

There is a range of researcher profiles available, but which are best for you?

It is highly recommended that RMIT researchers have an ORCID iD. Please see the section ORCID: a must have profile for more details.

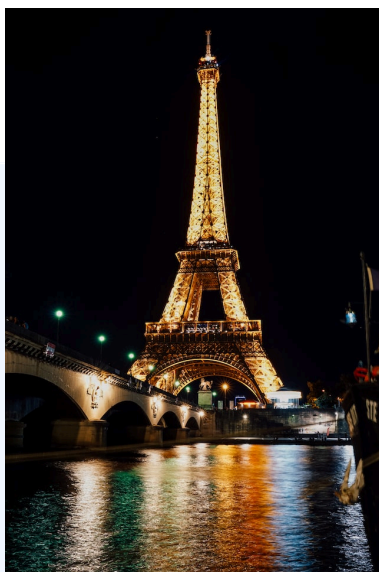


Photo by [Alex Azabache](#) on [Unsplash](#)

Beyond having an ORCID profile it can be confusing to decide which researcher profiles to use. Some considerations you might make include:

The purpose. Think about what you want from your profile. Are you using it to display your research outputs, calculate citations, network and share your work with other

researchers, identify collaborators or funding? If so, it may be worth maintaining a profile with Scopus or Web of Science (check if these index your papers first). For others it may be enough to have a Google Scholar profile.

Which profiles are relevant? How are other researchers in your discipline using their profiles – which are the best for your discipline?

Consider how many profiles you can regularly maintain. Don't feel compelled to have all profiles! All profiles require significant time to maintain.

Scopus Author ID

Scopus is a comprehensive bibliographic database that provides impact metrics for documents and authors, and enables the tracking of research over time. Your **Scopus Author ID is created automatically** when your first publication is indexed in the Scopus database. It allows you to see a list of your publications and view citation metrics. You can also link your Scopus Author ID to your ORCID profile.

Web of Science Researcher ID

Web of Science is another major citation database that

provides citation metrics for research publications and authors. A Web of Science ResearcherID allows you to claim and showcase your publications, and once created, the profile will assign a unique identifier known as the ResearcherID. The profile allows you to display and manage your affiliation, manage your publications, track citations and create citation reports, as well as display editing and peer review work for journals.

Google Scholar profile

Google Scholar is one of the major tools used to discover research publications. Setting up a Google Scholar Profile for yourself will make your publications easy to find.



Photo by Mitchell Luo on Unsplash

Google Scholar Citations allows you to create a profile and attach all of your Google Scholar publications and citations. It can also calculate your h-index and i10-index.

You can choose to make your profile public, so that it appears in Google Scholar results when people search for your name.

Comparing profiles

This table outlines some of the main features you may wish to consider when deciding which researcher profile you would like to use.

Researcher profiles comparison table

	ORCID	Google Scholar	Scopus	Web of Science
Publication list	Yes	Only those publications discoverable by Google Scholar	Only those publications included in the database	Only those publications included in the database
Non-Traditional Research Outputs (NTROs)	Yes	No	No	No
Citation metrics	No	Yes	Yes	Yes
Altmetrics	No	No	Some	No
Citation reports	No	No	Yes	Yes
Link with other profiles	Yes	No	Yes (ORCID)	Yes (ORCID)
Biography, affiliation, research interests	Yes	Yes	Affiliation only	Affiliation only
Awarded grants, invited positions	Yes	No	Some awarded grants	No
Link with publishers and funding bodies	Yes	No	No	No

Test your knowledge



An interactive H5P element has been excluded from this version of the text. You can view it online here:

<https://rmit.pressbooks.pub/researchwritingmodules/?p=2268#h5p-37>

BUILDING YOUR RESEARCHER PROFILES

Register for an ORCID iD

Researchers at RMIT University should use the RMIT Registration tool for an ORCID iD (login required).

Complete the required fields: name, primary email and password.

Choose a visibility setting for your ORCID profile, either public, limited or private. The default setting is public. You might, however, choose to change this to private until you have completed all of your profile details, such as employment and education, and added your citations.



Photo by [@charlybron](#) on Unsplash

Check to see if you have a Scopus Author ID

Go to Scopus (login required).

Do an Author Search on your name and your affiliation e.g. *RMIT University*. It may also be worth searching with previous affiliations. Did you find a record for yourself?

Make a note of your Author ID so that you can add this to your ORCID profile.

You can also get citation alerts and request corrections of your details in Scopus using this ID number. Further information on how to make corrections to your Scopus Author ID can be found in this [blog post](#).

Register for a Web of Science Researcher ID

First, to create a Web of Science researcher profile you will need to register for a Web of Science account at webofscience.com. Use your RMIT email address to create the account, and add an additional email address in the account settings.

After logging in to your Web of Science account, select the 'Researcher Profile' link on the left side navigation panel. This will display your researcher profile information, and will allow

you to claim and add publications, link with ORCID, create citation reports and manage your peer review activity.

Further information on how to make changes to your researcher profile can be found at the Web of Science help page.

Create a Google Scholar profile

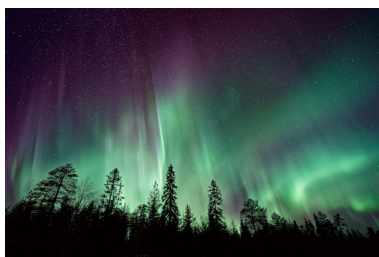
1. Go to Google Scholar Citations and sign in with your **private email account** details. Google recommends that you use a private email address when creating your profile. That way, your profile can travel with you if you leave RMIT.
2. Enter your name and your affiliation (i.e. RMIT University). **Make sure you enter your RMIT email address in the email verification field.** This will ensure that your profile is eligible for inclusion in search results. If you do move university you can easily update this address.
3. On the next page, add your publications to your profile.
4. Choose to have either automatic updates applied to your profile or to review them beforehand.
5. When satisfied with the results, make your profile public.

For more information see the [Google Scholar Citations help page](#).

CONNECTING EVERYTHING WITH ORCID

Connecting Scopus Author ID and ORCID

You can connect your Scopus Author ID to your ORCID so that your profile will be automatically updated with new publications.



Connecting Scopus Author ID and ORCID can

[Photo by Vincent Guth on Unsplash](#)

be done either in ORCID or in Scopus. You can choose either of the following options to link Scopus Author ID and ORCID.

Option 1: Link Scopus Author ID from your ORCID profile

1. Login to ORCID.
2. In your ORCID record, go to the Works section, 'Add works' and 'Search & link.'
3. Select Scopus Elsevier from the list and authorise.
4. Follow steps to associate your Scopus Author ID and works with ORCID.

Option 2: Link ORCID from your Scopus Author ID

1. Go to Scopus Author search (login required).
2. Search by your name and RMIT University in the Affiliation field.
3. Click on your name in the results list to open your profile page. (*Note* you won't find an Author ID if you don't have works already indexed by Scopus).
4. If you do have Author ID, Click on 'Add to ORCID' on the right-hand panel.
5. Follow the steps to associate your Scopus Author ID with your ORCID profile.

Connecting Web of Science

Researcher ID with ORCID

You might have associated your Web of Science Researcher ID with ORCID when you set it up. If not, you can do this at any point.

Note: that before you can do this your ORCID profile needs to be set to ‘public’.

Connect ResearcherID with ORCID

1. Log in to your Web of Science account.
2. Select ‘Settings’ then ‘Account Settings.’
3. In Account Settings, select the ‘Connected accounts’ tab, then select ‘Connect’ next to ORCID.

Enable syncing between Web of Science and ORCID

1. In the Web of Science Researcher ID menu, select ‘Edit’ next to your name.
2. Select the ‘ORCID Syncing’ tab and choose the options to:
 - ‘Update changes I make in my Web of Science profile’ and

- ‘Update my Web of Science Research Profile with changes I make in ORCID’.

Note: If you have undertaken peer reviews, you may want to select ‘Update ORCID with my peer reviews’.

Adding Google Scholar citations to ORCID

To add your articles from Google Scholar Citations into your ORCID profile you will need to create a BibTeX file in Google Scholar and import this into your ORCID record.

Follow these steps:

1. Go to your Google Scholar Citations profile page.
2. Select works to be exported.
3. Click export, select BibTeX and save the file.
4. Go to ORCID, login and go to your record.
5. Click Add Works and select Import BibTeX and select the file you saved in Google Scholar.

MANAGING YOUR RESEARCHER PROFILES

Managing your researcher profiles will enable you to effectively track engagement with your research outputs, and enable you to benchmark your performance against other researchers.



Photo by [Brett Jordan](#) on [Unsplash](#)

Once you have decided which profiles are suitable, and created and added information to those profiles, there are a few considerations to keep in mind.

Time commitment

Maintain your profiles regularly to ensure they are accurate, current and complete.

Risk management

Privacy – only share information you are happy for others to see.

Copyright and publishing agreements – check that you only share information that you can legally disclose to others.

Security – review the user agreement of each profile tool so that you know who accesses your data, how it is stored and what you can control.

Measuring impact

Citation metrics provided by the following researcher profiles are one way of measuring impact:

- Scopus Author ID
- Web of Sciences Researcher ID
- Google Scholar profile

These citation tools are limited as no single tool can provide a comprehensive measurement of research publication impact. They only gather metrics on the publications that they index. Citation counts should be used with other qualitative measures. **Altmetrics** (alternative metrics) provide insights into what is being said about research in non-scholarly forums (e.g. social media). Read more on the Altmetrics library guide.

Other key profile tools

The module has addressed the most common researcher profiles: ORCID, Scopus Author ID, Web of Science Researcher ID and Google Scholar profile.

You might consider other profiles tools:

- University profiles: RMIT staff profiles
- Social media: Twitter, LinkedIn, blogs
- Communities: Academia.edu, ResearchGate
- Dissemination: RMIT Research Repository, *The Conversation*, your email signature

There is much to consider when using social media. Engaging with social media enables you to promote and track your research to a wider audience. The Social Media for Researchers library guide introduces platforms and provides tips to maximise social media usage. Relevant university policies and guidelines, as well as copyright information are included.

Which researcher profiles will you use?

There are many considerations when deciding which researcher profiles will benefit you. Before creating any researcher profiles, consider:

- What is the purpose of the researcher profile and does it suit you?
- Which profiles are the most relevant for your research discipline?
- How many profiles can you regularly maintain?

EXPLORE FURTHER

Congratulations on completing this module. We have explored some common tools used to create author profiles and identifiers. These allow you to bring together and link your research outputs.

The module included setting up an ORCID iD, which acts as a universal identifier, and linking ORCID to other identifiers such as Scopus Author ID, Web of Science Researcher ID, and your Google Scholar author profile.



Photo by [Markus Spiske](#) on [Unsplash](#)

Further assistance

Individual assistance – For more assistance in setting up your researcher profiles contact the Library by submitting a Request Research Assistance form.

Library guide – Additional information is available from the online library guide Researcher Profiles and ORCID iDs.

Learn more

You may also be interested in viewing the webinar recording Bringing your researcher profiles together with ORCID (46 mins) (login required).

Feedback

Your comments and suggestions on how we can improve this module will be appreciated. Please complete our feedback form.

What's next?

You have completed this module on Setting Up a Researcher Profile. You can return to the Research and Writing Hub to choose another module to complete.

